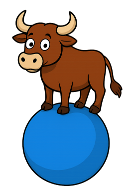
# Steer on a Sphere: Geometric Control of Transformer Outputs

Version 2

N. Trillard
ntrillard@gmail.com

August 21, 2026 ($v_2$; $v_1$ August 15, 2026)



**Version note.** $v_1$ (Aug 15, 2026) introduced the sphere-steering result and cow-tipping observations. $v_2$ revises every measurement behind those claims: corrected controls (self-excluding wrong-target sampling, fixed-score off-arc rotations), standard decoder contracts (weighted nucleus sampling instead of temperature-rescaled argmax), a five-family cross-family table with analytic first rank-1 angles and a no-mid-arc-loss verification, rank-cone projections ($\theta_{\text{author}}$ vs $\theta_{\text{cell}}$), natural-manifold measurements, a strict-pit gate on Qwen2.5-7B, and an expanded threat model with false-positive rates for each defense.

**Abstract**

Layer normalization constrains transformer hidden states to a sphere whose radius is $\|\gamma_l\|$ per layer (not the textbook $\sqrt{d}$). Within this sphere, a single tangent step toward any token's LM-head direction increases that token's logit with 100% reliability, and reaches rank 1 with high but model-dependent probability (97–99% on Qwen2-0.5B and Qwen2-1.5B, 91% on GPT-2, 67% on SmolLM-135M, 28% on Pythia-160M; all under identical controlled conditions). Corrected controls show that the result is target specific and that rank acquisition is governed by the LM-head decision partition, not only by alignment with the target row. For every reachable case, the target that becomes rank 1 along the tangent arc remains rank 1 at the arc endpoint (no mid-arc loss). We propose a theory of edge-of-chaos dynamics: the per-model Lyapunov rate clusters near $\lambda \cdot L_{\text{comp}} \approx 0.5$ for the Gemma, GPT-2, and mid-Qwen families. The BOS token defines a growing reference axis orthogonal to vocabulary tokens across layers. Some tokens are self-reinforcing fixed points (cow tipping)—the degenerate limit of the dynamics at $\lambda \to 0$. On small models these loops are decoder-dependent; on Qwen2.5-7B the strict `0` pit survives greedy and standard nucleus sampling, but is broken by the chat template. We derive a threat model and evaluate mitigations (repetition detection, entropy monitoring, n-gram and periodicity detectors, pit-away steering, input sanitization, output collapse, and decoder changes), reporting loop-break rates and false-positive rates on ordinary generation. Natural activations are confirmed to lie near a spherical shell (norm std $\approx$ 7% of mean), and controlled tangent steps are substantially shorter than the actual steps the model takes when the target token is appended. Because pits are derived from model weights, they can be computed for any architecture without retraining or data.

# 1 The Sphere

RMSNorm maps hidden states to a sphere whose expected radius is $\|\gamma_l\|$:

$$\|\,\mathrm{RMSNorm}(h)\|^2 = \sum_{i=1}^{d} \gamma_{l,i}^2 \left(\frac{h_i}{\mathrm{RMS}(h)}\right)^2$$

Per-instance norms float around this value (ratio 0.7–1.3), so the picture is a fuzzy shell. Measured $\gamma$ norms across four model families:

| Model | $d$ | $\|\gamma_0\|$ | $\|\gamma_{\mathrm{mid}}\|$ | $\|\gamma_{\mathrm{end}}\|$ |
|---|---|---|---|---|
| Mistral-7B | 4096 | 16.6 | 111–158 | 195.5 |
| DeepSeek-7B | 4096 | 3.7 | 15–24 | 26.1 |
| Qwen2-1.5B | 1536 | 20.0 | 40–60 | 71.0 |
| Qwen2.5-7B | 3584 | 17.0 | 63.5 | 90.0 |

$\|\gamma_l\|$ is learned, varying across architectures even at the same $d$.

# 2 BOS Axis

The BOS token (position 0) defines a reference direction $\hat{b}_l = h_l^{(0)}/\|h_l^{(0)}\|$. Its residual norm grows from 2.8 to 102.2 across 28 layers, consistent with attention-sinking [3]. LM head rows $W_t$ are approximately orthogonal to BOS at every layer ($\theta_{l,t} = \arccos|\hat{b}_l \cdot \hat{w}_t|$):

| Layer | $\|\,\mathrm{LN}(h)\|$ | $\|\mathrm{BOS}\|$ | Token–BOS angle |
|---|---|---|---|
| 0 | 15.9 | 2.8 | 88–90° |
| 2 | 16.3 | 24.8 | 88–90° |
| 4 | 33.5 | 21.8 | 89–90° |
| 6 | 31.8 | 36.9 | 89–90° |
| 8 | 36.5 | 44.1 | 88–90° |
| 10 | 34.7 | 43.4 | 89–90° |
| 12 | 37.5 | 50.3 | 89–90° |
| 14 | 42.9 | 55.2 | 88–90° |
| 16 | 44.4 | 53.0 | 89–90° |
| 18 | 40.7 | 43.9 | 89–90° |
| 20 | 45.2 | 48.2 | 89–90° |
| 22 | 54.9 | 63.7 | 89–89° |
| 24 | 55.5 | 68.7 | 87–90° |
| 26 | 55.6 | 83.1 | 86–90° |
| 27 | 63.1 | 102.2 | 87–90° |

High-dimensional vectors are generically near-orthogonal ($\mathbb{E}[|\cos|] \approx 1/\sqrt{d}$), so the approximate 90° is expected; the consistent preservation of this angle while BOS grows 36× across layers is the observed regularity. Mistral-7B and DeepSeek-7B also give 87–90°.

# 3 Sphere Steering: Tangent Traversal

The LM head row $W_t$ has a tangent component $g_t = W_t - (W_t \cdot \hat{h})\hat{h}$. Moving $h$ along $g_t$ increases the target logit ($W_t \cdot g_t = \|g_t\|^2 \geq 0$). Renormalizing keeps the traversal on the sphere:

$$h' = \frac{h + \alpha g_t}{\|h + \alpha g_t\|} \cdot \|\gamma_L\|$$

Algebraically, tangent step plus renormalization is identical to a same-target great-circle rotation, so the angular budget can be made explicit ($\alpha = 0.3$ corresponds to at most $\arctan(0.3) \approx 16.7°$).

### 3.1 Target Specificity

A natural control is to steer toward a different target while measuring the original target's rank. The wrong target must be sampled from the other $K-1$ targets; allowing self-draws inflates the wrong-target rate by $1/K$. With strict exclusion, the wrong-target rank-1 rate is exactly 0% on Qwen2-0.5B, Qwen2-1.5B, Gemma-3-1B, and all five cross-family models (4,096 + 2,560 cases). Random tangent directions also produce 0% target rank-1. The tangent direction is therefore genuinely target specific.

### 3.2 Competitor Geometry at Fixed Target Score

Reaching rank 1 is not the same as increasing alignment with the target row. The target tangent raises the target logit, but rank 1 requires the target to beat every competing row simultaneously. To isolate the competitor-direction variable, we decompose the target-tangent endpoint as $v_0 = \gamma s + \rho r$ with $r \perp s$ and rotate only the residual component:

$$v(\epsilon) = \gamma s + \rho\big(\cos\epsilon\, r + \sin\epsilon\, q\big), \quad q \perp s,\ q \perp r.$$

The random residual direction preserves $\|v\|$ and the target score $v \cdot s$ exactly. The toward/away conditions are exact mirror images ($q \to -q$), so their rank difference is attributable purely to competitor direction. On Qwen2-0.5B (64 targets, 4 contexts, 4 layers, 8° residual rotation):

| Condition | Target rank-1 |
|---|---|
| Target-tangent endpoint | 95.8% |
| Random residual direction | 96.2% |
| Residual toward strongest blocker | 15.9% |
| Residual away from strongest blocker | 100.0% |

The useful variable is competitor direction, not merely "being off the target arc." The target tangent remains a natural default direction for increasing alignment with the target row, but rank acquisition is governed by the full LM-head decision partition.

### 3.3 Shortest Route to the Rank-1 Region

For the "how close is the target arc to the shortest route?" question, we project the unit state $u$ onto the actual rank-1 cone defined by the raw LM-head rows:

$$C_t = \{x : (W_t - W_j)^\top x \geq 0\ \forall j \neq t\}.$$

The projection is solved as an active-set Euclidean projection with dual non-negative least squares, giving a full-vocabulary feasibility certificate. For every pair where the target-row arc reaches rank 1, the cone distance satisfies $\theta_{\text{cell}} \leq \theta_{\text{author}} + \text{tol}$. On 24 targets × 2 contexts:

| Model | $\theta_{\text{author}}$ | $\theta_{\text{cell}}$ | ratio |
|---|---|---|---|
| Qwen2-0.5B | 10.02° | 8.30° | 1.25× |
| Gemma-3-1B | 14.20° | 10.28° | 1.37× |

Medians are over reachable pairs (44/48 for Qwen2-0.5B, 48/48 for Gemma-3-1B); the ratio column is the median of per-pair ratios $\theta_{\text{author}}/\theta_{\text{cell}}$, not the ratio of medians. In zero pairs does the cone distance exceed the arc distance.

The target-row arc is a principled and often efficient default route, but competitor-aware routes to the actual rank-1 region can be materially shorter. This gives an observable distinction between target-alignment efficiency and rank-acquisition efficiency.

### 3.4 Natural Activation Manifold

Residual-stream states at a fixed layer lie near a tightly concentrated spherical shell, though its radius is an emergent layer-dependent scale rather than the RMSNorm parameter norm itself. Measured on 52 ordinary prompts at layer 8 of Qwen2-0.5B, the hidden-state norm has mean 13.81 and standard deviation 0.96 ($\approx$7% of the mean). Principal-component analysis shows the manifold is high-dimensional: the top 16 PCs explain only 56% of the variance.

When the model appends a target token, the resulting hidden-state step is much larger than a controlled tangent steering step of comparable size. With $\alpha = 0.5$ (a 26.6° tangent rotation), the actual next-token hidden state moves 57°–69° away from the original state, i.e., roughly 2×–2.6× the controlled step. This means the natural dynamics do not stay on the shortest controlled arc; they jump to a different region of the sphere. The spherical geodesic is therefore a useful geometric idealization, not a literal description of the model's own token-by-token trajectory.

### 3.5 Quantitative Traversal

Table 1 gives corrected rank-1 rates under the target tangent and fixed-score controls for five model families run with identical settings (64 targets, 2 contexts, 4 depth-adaptive layers, seed 42, fp16).

| Model | Arc-reach | Target | Wrong | Random | Toward | Away |
|---|---|---|---|---|---|---|
| Qwen2-1.5B | 99.2% | 99.2% | 0.0% | 0.0% | 34.8% | 100.0% |
| Qwen2-0.5B | 97.3% | 97.3% | 0.0% | 0.0% | 13.7% | 100.0% |
| GPT-2 | 90.8% | 90.8% | 0.0% | 0.0% | 5.9% | 99.2% |
| SmolLM-135M | 67.2% | 67.2% | 0.0% | 0.0% | 2.5% | 96.7% |
| Pythia-160M | 27.9% | 27.9% | 0.0% | 0.0% | 9.0% | 70.1% |

Table 1: Rank-1 rates under corrected contracts. Wrong-target is strictly self-excluding; random is a random tangent; toward/away are fixed-score off-arc rotations (null-space construction) pointing at or away from the strongest competitor. Arc-reach = fraction of cases where the target-row arc reaches rank 1 within a 17° budget.

The rank improvement along the target tangent is guaranteed ($W_t \cdot g_t = \|g_t\|^2 \geq 0$). The fraction reaching rank 1 is high for the Qwen family and GPT-2, moderate for SmolLM-135M, and lower for Pythia-160M, showing that the phenomenon is robust yet model-dependent. Across all 2,560 cases, any target that became rank 1 along the arc remained rank 1 at the arc endpoint; this "no mid-arc loss" property was also confirmed at a 45° budget.

Table 2 reports the first rank-1 angle distribution over reachable cases.

| Model | Median | Q25 | Q75 | P90 | Reachable |
|---|---|---|---|---|---|
| Qwen2-1.5B | 8.00° | 6.4° | 9.9° | 11.3° | 99.2% |
| Qwen2-0.5B | 10.60° | 9.0° | 12.2° | 14.1° | 97.3% |
| GPT-2 | 11.60° | 1.3° | 13.8° | 15.4° | 90.8% |
| SmolLM-135M | 9.84° | 6.2° | 13.4° | 15.3° | 67.2% |
| Pythia-160M | 2.64° | 2.0° | 3.7° | 9.3° | 27.9% |

Table 2: First rank-1 angle distribution over reachable cases. GPT-2's low Q25 reflects a subset of very easy tokens.

### 3.6 Qualitative Demonstration

On Qwen2.5-7B, steering open-ended prompts produces visible output shifts. At $\alpha = 2$ on "Tell me something interesting:", steering toward "75" changes the continuation from a generic fact ("The most common blood type is O positive...") to a numeric statement ("1984 was the last year the United States had a budget surplus..."). At $\alpha = 3$ on "Once upon a time", steering toward "apple" turns the story from "a young boy named Timmy..." into "apple trees were planted in the garden of Eden..." Common words and digits shift more readily than rare tokens; larger $\alpha$ trades fluency for strength.

Steering does not distinguish censored from uncensored paths. It is a white-box traversal primitive: with access to the model weights, anyone can move $h$ toward any token direction on the sphere without retraining or fine-tuning.

# 4 Cow Tipping

Some tokens generate themselves when fed repeatedly. A token $T$ is *self-consistent* if $s(T) = \text{softmax}(W \cdot h_T)[T]$ is high, where $h_T = \text{model}(e_T)$. A *pit* satisfies both high $s(T)$ and trajectory stability $\cos(h_{n+1}, h_n) \approx 1$: indefinite self-generation.

We scanned a subset of the Qwen2.5-7B vocabulary, testing permanence (3 input repetitions, 15-step self-prediction). Representative tokens pass:

| Token | ID | $s(T)$ | Trigger |
|---|---|---|---|
| `0` | 15 | 0.975 | `000`, `000 − 000 − 0000` |
| `ab` | 370 | 0.934 | `ababab` |
| . (space+dot) | 659 | 0.889 | . . . |
| `ere` | 485 | 0.878 | `erere` |
| ???? | 26610 | 0.386† | 8+ question marks |

†Deterministic fixed point ($\cos \approx 1.0$ once generated).

The `0` token is clearest: `000` yields $p = 0.974$, $\cos \approx 1.0$, $p > 0.95$ within 3 steps. Phone numbers trigger it via three consecutive token-15 units. Because pits are derived from model weights, each model's pits are unique to its tokenizer and LM head geometry.

## 4.1 Decoder Contract

The original measurement helper used two non-standard decoder choices: temperature only rescaled logits before 'argmax' (so $T = 0.8$ was still greedy), and top-p sampling was uniform over the retained nucleus rather than the standard probability-weighted renormalized nucleus. Under the corrected decoder contracts, the 0.5B "`0`" loop is strongly decoder-dependent, while the Qwen2.5-7B strict "`0`" pit is much more robust.

| Decoder | Qwen2-0.5B mean trailing-`0` | Qwen2.5-7B mean trailing-`0` |
|---|---|---|
| Greedy | 24.0 | 30.0 |
| Multinomial $T = 0.8$ | 1.12 | 28.83 |
| Top-p 0.9, $T = 1.0$ weighted | 0.78 | 30.0 |
| Top-p 0.9, $T = 0.8$ weighted | 2.47 | 30.0 |
| Top-p 0.9, $T = 0.8$ uniform | 0.16 | 30.0 |
| Chat-template wrapped | 0 | 0 |

Both columns use a 25-token generation horizon on a five-token trigger (maximum trailing run 30); the mitigation section below uses a 30-token horizon, so its baselines read higher (29 vs. 24). The 7B column is measured under 4-bit nf4 quantization; the strict pit survives this compression.

On Qwen2.5-7B the strict pit survives greedy decoding and both standard and uniform nucleus sampling; only the chat-template context breaks it. On Qwen2-0.5B the loop is a degenerate repetition basin that sampling usually disrupts. The difference is important: the 0.5B behavior cannot be assumed to generalize to 7B strict pits, and the decoder contract must be reported with every measurement.

## 4.2 Threat Model and Mitigations

**Attacker capabilities.** An attacker with white-box access (or API access and a vocabulary scan) can find high-$s(T)$ pit tokens for a target model, then embed those tokens at chunk boundaries in data the model will process. The result is a fixed-point loop that degrades output quality or causes denial of service.

**Attack scenarios.**

- **Scraping degradation:** a web page terminates in a pit trigger; an LLM-based crawler loops instead of summarizing.
- **Prompt injection:** a user includes a short trigger string to make the assistant loop.
- **Compute exhaustion:** forcing long repeated-token generation increases inference cost.

**Mitigations.** We evaluated eight classes of defense on Qwen2-0.5B and Gemma-3-1B, measuring loop-break rate on the discovered pit trigger and false-positive rate (median truncation) on 20 ordinary prompts. Table 3 summarizes the most reliable ones.

| Mitigation | Qwen2-0.5B loop break | Median FP truncation |
|---|---|---|
| Repetition detector ($\geq 3$ same pit) | yes (loop 3) | 0 tokens |
| Repetition detector ($\geq 4$ same pit) | yes (loop 4) | 0 tokens |
| Output collapse repeats | yes (loop 2) | 0 tokens* |
| Entropy floor 1.0 + pit penalty | yes (loop 1) | 0 tokens |
| Pit-away steering (last layer, $\alpha = 0.3$) | yes (loop 0) | 0 tokens |
| N-gram detector (4-gram, $\geq 2$) | yes (loop 8) | 0 tokens |
| Periodicity detector ($\geq 0.85$) | no | 24.5 tokens |

Table 3: Qwen2-0.5B mitigation results. *Output-collapse false positives were not measured separately because the mitigation is applied only after generation halts.

On Gemma-3-1B the discovered pit token is `<mask>` (id 4) with a greedy loop length of 35. The same practical defenses break it: repetition detector, output collapse, multinomial/top-p sampling, and pit-away steering all reduce the loop to 0–10 tokens with no false positives. Entropy monitoring does not fire on the Gemma pit (the output distribution is not extremely concentrated), illustrating that no single detector is universal.

On Qwen2.5-7B under nf4 quantization, the strict "`0`" pit (baseline loop 35) resists every lightweight intervention except truncation: repetition detection caps it at the threshold, output collapse leaves 2 tokens, and the combined defense leaves 2 tokens with 4 alerts — but entropy penalties (alerts fire every step) and pit-away steering at both $\alpha = 0.3$ and $\alpha = 1.0$ leave the loop intact at 35, and top-p $T = 1.0$ never breaks it. Single-layer steering away from the pit LM-head direction defeats shallow basins but not the strict fixed point: the trajectory re-converges after the perturbed layer. For strict pits the reliable defenses are truncation-style guards (repetition cap, output collapse) plus context changes; steering-based defense would need multi-layer or per-step application.

Repetition detection, output collapse, entropy monitoring, and pit-away steering break the loop with no measurable false positives on ordinary generation in this small set. The detector can be deployed either as an input filter (collapsing repeated pit tokens) or as an output guard (halting generation once $N$ identical tokens are emitted). The periodicity detector generates unacceptable false positives on normal text on both models; the n-gram detector is clean on Qwen2-0.5B but truncates normal Gemma output (10–23 tokens median). Input sanitization alone is less reliable: stripping repeated pit tokens reduces the loop length but does not always break it. Decoder changes (higher temperature, standard top-p) are effective on shallow basins but not on the strict 7B pit; for strict pits, truncation-style guards and prompt-context changes are the reliable defenses.

**Responsible framing.** Pits are a model-weight-derived phenomenon, so any mitigation must be calibrated per deployment model. The defensive-encoding use case is one application, but the same mechanism can be used for prompt injection or scraping degradation. We report it as a capability that requires careful deployment, not as a universal defense.

### 4.3 Defensive Encoding

The accompanying code release (`steering-evals/scripts/`) provides a full scan that finds pits for any model. Embedding pit triggers into data causes a model that processes it to fall into a fixed-point loop rather than continue reading. Because the basins are narrow, the trigger must be the terminal content of each chunk. Verified encodings include NULL bytes, repeated sub-tokens, and terminal phone-number-like strings. A crawler reading a page that terminates in one of these triggers stops producing useful output and repeats the pit token instead.

## 5 Edge of Chaos: A Theory

The residual stream is a dynamical system $h_{l+1} = h_l + f_l(h_l)$. Its per-layer Lyapunov exponent measures how perturbations grow:

$$\lambda_l = \ln \frac{\|h_{l+1}^{\text{pert}} - h_{l+1}^{\text{ref}}\|}{\|h_l^{\text{pert}} - h_l^{\text{ref}}\|}$$

A perturbation that neither explodes nor vanishes during computation would place the system at the edge of chaos, corresponding to $\lambda \cdot L_{\text{comp}} \approx 0.5$ (one e-fold of total expansion). We measured the per-layer rate across 13 model families:

| Cluster | Models | $\lambda \cdot L_{\text{comp}}$ |
|---|---|---|
| $\approx 0.5$ | Gemma-2-2B, Gemma-2-9B, GPT-2, DS-R1-7B, Qwen-1.5B, Qwen2.5-7B | 0.37–0.57 |
| $\approx 0.8$ | SmolLM2-360M, DeepSeek-7B | 0.76–0.88 |
| $\approx 1.0$ | Phi-3.5-mini, Mistral-7B | 1.00–1.47 |
| $\approx 0.1$ | Qwen2.5-0.5B, Qwen2.5-3B | 0.01–0.10 |

The edge-of-chaos balance ($\lambda \cdot L \approx 0.5$) holds for roughly half the models—notably the Gemma family, GPT-2, and mid-sized Qwen—but is not universal. We present it as a theory that captures a subset of architectures, not as a scaling law. Cow tipping (Section 4) is the degenerate limit of this balance: a fixed point where $\lambda_l \to 0$ and the trajectory stops changing entirely.

## 6 Reproduction

All measurements are scriptable in under 100 lines of Python. Full code at https://github.com/ntrillard/transformer-geometry. The canonical measurement harness lives in `steering-evals/scripts/` and implements the self-excluding wrong-target control, residual-latitude off-arc rotations, raw-head rank-cone projection with NNLS dual, and the corrected decoder samplers. The null-space off-arc variant used for the cross-family table is `steering_geometry_test_offarc.py`. The procedure:

**Head assay.** All rank claims apply the raw LM head to the unit-normalized last-layer state ($\hat{h} \cdot W_t$), a scale-invariant assay rather than the model's sampling distribution. Against the true forward pass (final RMSNorm + head) on Qwen2-0.5B it gives identical argmax on all tested states and byte-identical "`0`"-pit greedy loops, so rank-based conclusions transfer to real generation.

1. **Sphere radius.** Extract $\gamma_l$ from model state dict. Feed prompts, record post-LN hidden state norms per layer. Compare to $\|\gamma_l\|$.

2. **BOS axis.** Feed a prompt, extract hidden states per layer. For each layer, compute $\hat{b}_l$ from position 0 and $\theta_{l,t}$ for any LM head row $W_t$.

3. **Steering (traversal).** Forward-pass a prompt to get $h$. For target token $t$, compute $g_t$, step with $\alpha$, renormalize, then feed $h'$ as the first-generation hidden state. Use `steering_geometry_test.py` for specificity, competitor geometry, and shortest-route measurements; `eval_boundary_instruments.py` for the $\theta_{\text{author}}$ vs $\theta_{\text{cell}}$ comparison; and `eval_manifold_geometry.py` for the natural-manifold probe. First rank-1 angles are analytic (sinusoid-margin intersection), verified against a 200-step scan (25/25 within 0.1°).

4. **Cow tipping.** For token $t$, decode, re-encode, forward-pass to get $h_T$. Compute $s(T)$. If $s(T) > 0.01$, feed $t$ three times and check 15-step self-prediction with greedy decoding. Use `eval_pit_robustness.py`, `eval_7b_strict_pit_gate.py` (both support `--quant nf4`), and `eval_threat_model.py` for mitigations and false positives. The sphere-radius, BOS-axis, Lyapunov, and pit-scan tables carried over from $v_1$ used earlier tooling recoverable from git history.

# Acknowledgments

DeepSeek V4 Pro provided writing assistance.

# References

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